Luis Enrique Dalmau Coopat

Full-Stack AI Engineer · Computer Vision · YOLOv9

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SUMMARY

Full-Stack AI Engineer with **3+ years** building production systems across computer vision, ML pipelines, and full-stack web applications. Author of a proprietary computer-vision library used to train and deploy **YOLOv9** object-detection models for industrial IIoT predictive maintenance. B.S. in Computer Science, hands-on MLOps, and a teaching background in AI, ML, and distributed systems. Bilingual Spanish/English.

SKILLS

AI & Computer Vision: YOLOv9 · PyTorch · OpenCV · Object Detection · Image Segmentation · Auto-labeling (GroundingDINO, SAM, Florence-2) · Synthetic Data (Stable Diffusion) · Albumentations · TensorBoard · scikit-learn · pandas · NumPy
Backend: FastAPI · SQLAlchemy 2.0 · Alembic · Django · Pydantic · REST APIs · JWT/OAuth2 · APScheduler
Frontend: React 18 · Next.js 14 · TypeScript · Tailwind CSS · TanStack Query · Formik · Chart.js
Databases: PostgreSQL · MongoDB · MySQL · Redis
DevOps & Cloud: Docker · Docker Compose · Heroku · AWS S3 · Nginx · Git · GitLab CI · Linux
Testing & Quality: pytest · pytest-asyncio · pytest-cov · moto · httpx · ESLint · Prettier

EXPERIENCE

Technical Engineer
10/2025 – Present

Anukys · Malaga, Spain (Hybrid)

- Consultancy serving clients and business partners in artificial intelligence and computer vision.

AI Engineer — Computer Vision
01/2024 – Present

SORBA.ai (Sorbotics LLC) · Jacksonville, USA (Remote)

- Built and optimized CV training pipelines covering 10+ architectures (YOLOv5/v9, EfficientDet, MobileNet v2, RetinaNet, Faster R-CNN, Mask R-CNN, CenterNet, SSD).
- Shipped these capabilities across 22 company projects, from dataset preparation to on-plant deployment.
- Integrated synthetic data generation (Stable Diffusion, GAN, VAE), auto-labeling (GroundingDINO, Florence-2), and model optimization (TensorRT, TFLite).
- Built real-time video processing for industrial detection: fire/smoke, gas leaks, PPE, gauge OCR, license plates, and spatial tracking.
- Packaged ML models for Debian/RHEL. Managed GPU resources for concurrent training.

Full-Stack Developer (Backend focus)
09/2024 – 10/2025

Aerial Underground Construction Inc. · Kissimmee, FL, USA (Remote)

- Built logistics platform backend (FastAPI + SQLAlchemy + Alembic + PostgreSQL): JWT auth, RBAC (3 roles), 20+ REST endpoints.
- Implemented commissions, daily/weekly/monthly reports, PDF/Excel invoices (ReportLab/openpyxl), AWS S3 (boto3), scheduled jobs (APScheduler).
- Delivered ~35% of a Next.js 14 + React + TypeScript frontend with automated tests and Docker/Heroku deployment.

Assistant Professor
01/2023 – 01/2025

University of Havana · Faculty of Mathematics & Computer Science

- Taught Computer Networks, Machine Learning, AI, Operating Systems, and Distributed Systems; led practicals, graded exams, mentored students.

SELECTED PROJECTS

Computer Vision Library (proprietary) — author: YOLOv9 training, auto-labeling, synthetic data, GPU management.

Logistics Management System — FastAPI + Next.js + PostgreSQL; RBAC, reports, invoices, AWS S3.

YOLO-3D — Real-time 3D object detection: YOLOv11 + Depth Anything v2.

Image Analyzer (BLIP) — Multimodal image analysis: descriptions, keywords.

LicitAI — FastAPI + React + MongoDB + Google Gemini (LLM) + Stripe.

Protectora RRSS — Pet-adoption diffusion: FastAPI + React + Telegram bot + SMTP.

ML Trainer Engine — CV training with SOTA models (RF-DETR, EfficientDet, MobileNet, DeepLabV3).

EDUCATION & CERTIFICATIONS

B.S. in Computer Science — University of Havana (2017–2022). Thesis: "Graphic generation of signs of the Cuban Sign Language".

CS50's Introduction to AI with Python — HarvardX (edX) · **Data Science & Machine Learning** — Sololearn.

LANGUAGES

Spanish: Native · **English:** B2 (Professional Working Proficiency)